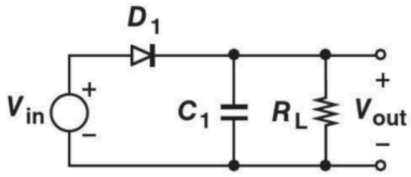
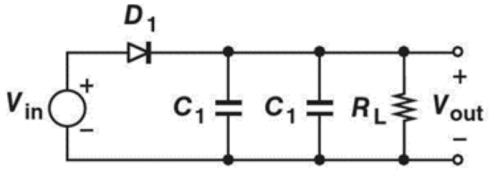
Midterm

5 Problems (20 points each, total 100 points)

1. The half-wave rectifier shown below drives a 100-Ω load through a diode with $V_{D,on} = 0.6\,\text{V}$ and a 500-μF smoothing capacitor as shown in (a). The input is a 60-Hz sinusoidal voltage source with a peak voltage of 3.5 V. If the circuit is replaced by (b), Calculate the ripple amplitude respectively (20 points)



(a)



(b)

$V_{out,\text{peak}} = V_p - V_{D,on} = 3.5 - 0.6 = 2.9\ \text{V}$

$$I_L = \frac{V_{out,\text{peak}}}{R_L} = \frac{2.9\ \text{V}}{100\ \Omega} = 29\ \text{mA}$$

$$V_R \approx \frac{I_L}{C\,f_{in}}$$

(a) $$V_R = \frac{29 \times 10^{-3}}{(500 \times 10^{-6})(60)} = 0.967\ \text{V}$$

(b) $C_{eq} = C_1 + C_1 = 1000\ \mu\text{F} \Rightarrow$ $$V_R = \frac{29 \times 10^{-3}}{(1000 \times 10^{-6})(60)} = 0.483\ \text{V}$$

Ans. (a) $V_R = 0.967\ \text{V}$ (b) $V_R = 0.483\ \text{V}$

2. A junction employs $N_D = 4 \times 10^{16}\ \text{cm}^{-3}$ and $N_A = 3 \times 10^{15}\text{cm}^{-3}$ Calculate the built-in potential at (a) T = 300 K and (b) 350 K. (20 points)

$$V_0 = V_T \ln \frac{N_A N_D}{n_i^2}, \qquad V_T = \frac{kT}{q}, \qquad n_i = 5.2 \times 10^{15}\, T^{3/2} e^{-E_g/(2kT)}$$

(a) $T = 300\ \text{K}:\quad V_T = 25.9\ \text{mV}, \quad n_i = 1.08 \times 10^{10}\ \text{cm}^{-3}$

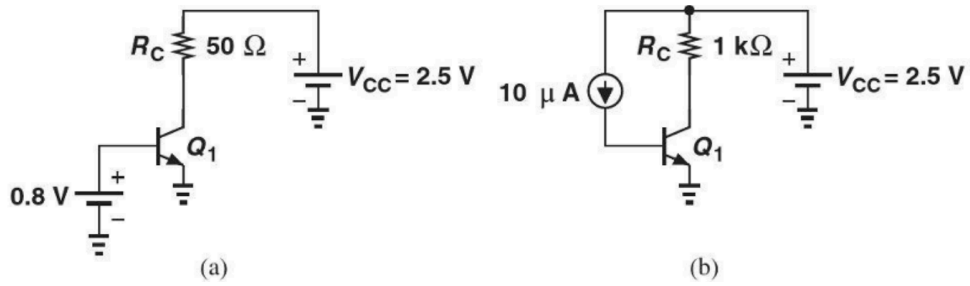
$$V_0 = (25.9\ \text{mV}) \ln \frac{(3 \times 10^{15})(4 \times 10^{16})}{(1.08 \times 10^{10})^2} = 0.716\ \text{V}$$

(b) $T = 350\ \text{K}:\quad V_T = 30.2\ \text{mV}, \quad n_i = 2.99 \times 10^{11}\ \text{cm}^{-3}$

$$V_0 = (30.2\ \text{mV}) \ln \frac{(3 \times 10^{15})(4 \times 10^{16})}{(2.99 \times 10^{11})^2} = 0.635\ \text{V}$$

Ans. (a) $V_0 = 0.716\ \text{V}$ (b) $V_0 = 0.635\ \text{V}$

3. Determine the operating point and the small-signal model of Q for the circuits shown in (a) and (b). Assume $I_S = 2\text{x}10^{-16}$ A, $\beta = 100$, and $V_A = \infty$. (20points)



(a) $V_{BE} = 0.8$ V

$$I_C = I_S e^{V_{BE}/V_T} = (2 \times 10^{-16}) e^{0.8/0.026} = 4.61 \text{ mA}$$

$$I_B = \frac{I_C}{\beta} = 46.1\ \mu\text{A}, \qquad I_E = I_C + I_B = 4.66 \text{ mA}$$

$$V_{CE} = V_{CC} - I_C R_C = 2.5 - (4.61 \text{ mA})(50\ \Omega) = 2.27 \text{ V}$$

$$g_m = \frac{I_C}{V_T} = 177 \text{ mS}, \qquad r_\pi = \frac{\beta}{g_m} = 564\ \Omega, \qquad r_o = \frac{V_A}{I_C} = \infty$$

(b) $I_B = 10\ \mu\text{A}$

$$I_C = \beta I_B = 1 \text{ mA}, \qquad I_E = 1.01 \text{ mA}, \qquad V_{CE} = 2.5 - (1 \text{ mA})(1 \text{ k}\Omega) = 1.5 \text{ V}$$

$$V_{BE} = V_T \ln \frac{I_C}{I_S} = (26 \text{ mV}) \ln \frac{10^{-3}}{2 \times 10^{-16}} = 760 \text{ mV}$$

$$g_m = \frac{I_C}{V_T} = 38.5 \text{ mS}, \qquad r_\pi = \frac{\beta}{g_m} = 2.6 \text{ k}\Omega, \qquad r_o = \infty$$

Ans. (a) $I_C = 4.61$ mA, $I_B = 46.1\ \mu$A, $V_{CE} = 2.27$ V; $g_m = 177$ mS, $r_\pi = 564\ \Omega$, $r_o = \infty$
(b) $I_C = 1$ mA, $V_{BE} = 760$ mV, $V_{CE} = 1.5$ V; $g_m = 38.5$ mS, $r_\pi = 2.6$ kΩ, $r_o = \infty$

4. A bipolar transistor carries a collector current of 1mA with $V_{CE} = 2V$. Determine the required base-emitter voltage if (a) $V_A = \infty$ or (b) $V_A = 20V$. Assume $I_S = 3 \times 10^{-16}$ A.

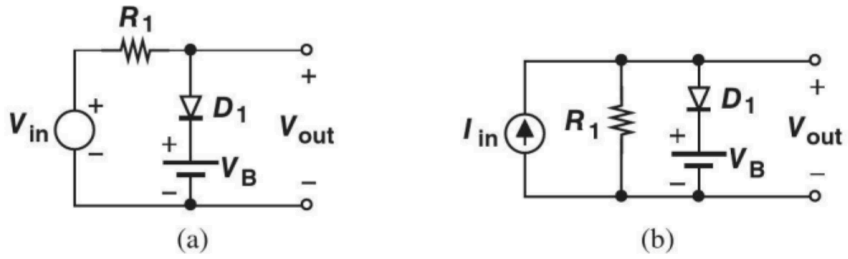
$$I_C = I_S e^{V_{BE}/V_T}\left(1 + \frac{V_{CE}}{V_A}\right) \Rightarrow V_{BE} = V_T \ln\left(\frac{I_C}{I_S} \cdot \frac{1}{1 + V_{CE}/V_A}\right)$$

(a) $V_A = \infty$: $\quad V_{BE} = (26\ \text{mV}) \ln \dfrac{10^{-3}}{3 \times 10^{-16}} = 750\ \text{mV}$

(b) $V_A = 20\ \text{V}$: $\quad 1 + \dfrac{2}{20} = 1.1, \qquad V_{BE} = (26\ \text{mV}) \ln\left(\dfrac{10^{-3}}{3 \times 10^{-16}} \cdot \dfrac{1}{1.1}\right) = 747\ \text{mV}$

Ans. (a) $V_{BE} = 750\ \text{mV}$ $\quad$ (b) $V_{BE} = 747\ \text{mV}$

5. In the circuits of (a) and (b), plot the current flowing through R_1 as a function of V_{in} *or* I_{in}. Assume a constant-voltage diode model with $V_{Don} = 0.8V$ and $V_B = 1V$ (20points)



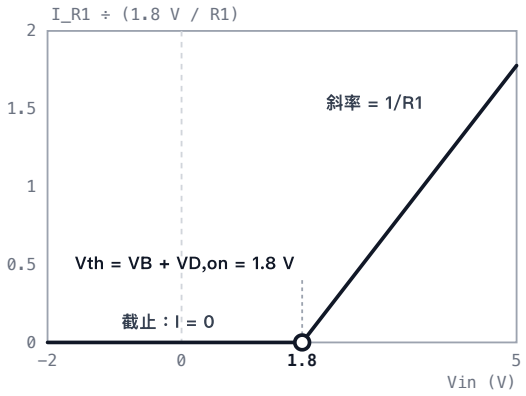
(a) (b)

D_1 on : $V_{out} \geq V_B + V_{D,on} = 1 + 0.8 = 1.8\ \text{V}$

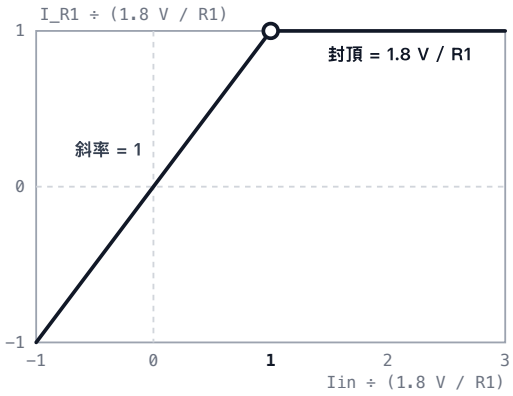
(a) $V_{in} < 1.8\ \text{V}: I_{R_1} = 0;$ $V_{in} \geq 1.8\ \text{V}: I_{R_1} = \dfrac{V_{in} - 1.8}{R_1}$

(b) $I_{in} < \dfrac{1.8}{R_1}: I_{R_1} = I_{in};$ $I_{in} \geq \dfrac{1.8}{R_1}: I_{R_1} = \dfrac{1.8}{R_1}$

(a) 電壓驅動：I_{R1} 對 V_{in}



(b) 電流驅動：I_{R1} 對 I_{in}



Ans. (a) 轉折於 $V_{in} = 1.8\ \text{V}$，之後斜率 $1/R_1$　　(b) 轉折於 $I_{in} = 1.8/R_1$，之後定值 $1.8/R_1$